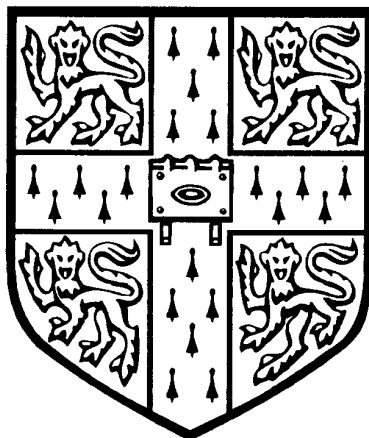


List MF7 (To be used in and after March 1996)
For Advanced Level/HSC Principal

Mathematics and Further Mathematics

University of Cambridge Local Examinations Syndicate



LIST OF FORMULAE

AND

STATISTICAL TABLES

$$\begin{aligned} & 3.13 \times 2.7 + 2.5 \times 1.7 \\ & = 8.451 + 4.25 \\ & = 12.701 \end{aligned}$$

PURE MATHEMATICS

Algebra

For the quadratic equation $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

For an arithmetic series:

$$u_n = a + (n - 1)d, \quad S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n - 1)d\}$$

For a geometric series:

$$u_n = ar^{n-1}, \quad S_n = \frac{a(1 - r^n)}{1 - r} \quad (r \neq 1)$$

Binomial expansion:

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1}b + \binom{n}{2} a^{n-2}b^2 + \dots + b^n, \text{ where } n \text{ is a positive integer}$$

Summations:

$$\sum_{r=1}^n r = \frac{1}{2}n(n + 1), \quad \sum_{r=1}^n r^2 = \frac{1}{6}n(n + 1)(2n + 1), \quad \sum_{r=1}^n r^3 = \frac{1}{4}n^2(n + 1)^2$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area of triangle} = \frac{1}{2}ab \sin C$$

$$\text{Arc length} = r\theta \quad (\theta \text{ in radians})$$

$$\text{Area of sector} = \frac{1}{2}r^2\theta \quad (\theta \text{ in radians})$$

$$\cos^2 \theta + \sin^2 \theta \equiv 1, \quad 1 + \tan^2 \theta \equiv \sec^2 \theta, \quad \cot^2 \theta + 1 \equiv \operatorname{cosec}^2 \theta$$

$$\sin(A \pm B) \equiv \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) \equiv \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) \equiv \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\cos 2A \equiv \cos^2 A - \sin^2 A \equiv 2 \cos^2 A - 1 \equiv 1 - 2 \sin^2 A$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

Principal values:

$$-\frac{1}{2}\pi \leq \sin^{-1} x \leq \frac{1}{2}\pi$$

$$0 \leq \cos^{-1} x \leq \pi$$

$$-\frac{1}{2}\pi < \tan^{-1} x < \frac{1}{2}\pi$$

Hyperbolic functions

$$\cosh x \equiv \frac{1}{2}(e^x + e^{-x}), \quad \sinh x \equiv \frac{1}{2}(e^x - e^{-x})$$

$$\cosh^2 x - \sinh^2 x \equiv 1$$

$$\sinh 2x \equiv 2 \sinh x \cosh x$$

$$\cosh 2x \equiv \cosh^2 x + \sinh^2 x$$

$$\sinh^{-1} x \equiv \ln\{x + \sqrt{(x^2 + 1)}\}$$

$$\cosh^{-1} x \equiv \ln\{x + \sqrt{(x^2 - 1)}\} \quad (|x| \geq 1)$$

$$\tanh^{-1} x \equiv \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad (|x| < 1)$$

Differentiation

$f(x)$	$f'(x)$
x^n	nx^{n-1}
$\ln x$	$\frac{1}{x}$
e^x	e^x
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$
$\sinh^{-1} x$	$\frac{1}{\sqrt{1+x^2}}$
$\cosh^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$
$\tanh^{-1} x$	$\frac{1}{1-x^2}$
uv	$v \frac{du}{dx} + u \frac{dv}{dx}$
$\frac{u}{v}$	$\frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $x = f(t)$ and $y = g(t)$ then $\frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt}$

Integration

$f(x)$	$\int f(x) dx$
x^n	$\frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$
$\frac{1}{x}$	$\ln x + c$
e^x	$e^x + c$
$\sin x$	$-\cos x + c$
$\cos x$	$\sin x + c$
$\sec^2 x$	$\tan x + c$
$\frac{1}{\sqrt{a^2-x^2}}$	$\sin^{-1} \left(\frac{x}{a} \right) + c$
$\frac{1}{a^2+x^2}$	$\frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$
$\frac{1}{\sqrt{x^2-a^2}}$	$\cosh^{-1} \left(\frac{x}{a} \right) + c$
$\frac{1}{\sqrt{x^2+a^2}}$	$\sinh^{-1} \left(\frac{x}{a} \right) + c$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

Arc length:

$$\int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{or} \quad \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Surface area of revolution:

$$S_x = 2\pi \int_{x=x_1}^{x=x_2} y ds, \quad S_y = 2\pi \int_{y=y_1}^{y=y_2} x ds$$

Series expansions

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots \quad (|x| < 1, n \text{ rational})$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad \text{for all } x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \quad \text{for all } x$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots, \quad \text{for all } x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots, \quad \text{for } -1 < x \leq 1$$

Differential Equations

An integrating factor for $\frac{dy}{dx} + Py = Q$ is $e^{\int P dx}$

Complex numbers

De Moivre's theorem:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

Vectors

For $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$:

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

Numerical methods

The Newton-Raphson iteration for solving $f(x) = 0$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Trapezium Rule:

$$\int_a^b f(x) dx \approx \frac{1}{2}h\{y_0 + 2(y_1 + y_2 + \dots + y_{n-1}) + y_n\}$$

MECHANICS

Uniformly accelerated motion

$$v = u + at,$$

$$s = \frac{1}{2}(u + v)t,$$

$$s = ut + \frac{1}{2}at^2,$$

$$v^2 = u^2 + 2as$$

Centres of mass of uniform bodies

Triangular lamina: $\frac{2}{3}$ along median from vertex

Solid hemisphere: $\frac{3}{8}r$ from centre

Hemispherical shell: $\frac{1}{2}r$ from centre

Circular arc of angle 2α : $\frac{r \sin \alpha}{\alpha}$ from centre

Circular sector of angle 2α : $\frac{2r \sin \alpha}{3\alpha}$ from centre

Solid cone: $\frac{3}{4}h$ from vertex

Elastic strings and springs

Hooke's law:

$$T = \frac{\lambda x}{l}$$

Circular motion

For uniform circular motion, the acceleration is directed towards the centre and has magnitude

$$\omega^2 r \quad \text{or} \quad \frac{v^2}{r}$$

Simple harmonic motion

$$\text{Period} = \frac{2\pi}{\omega}$$

$$v^2 = \omega^2(a^2 - x^2)$$

$$x = a \sin \omega t \quad \text{or} \quad x = a \cos \omega t \quad \text{or, in general,} \quad x = a \cos(\omega t + \epsilon)$$

Moments of inertia

Uniform rod, length $2a$, about perpendicular axis through centre: $\frac{1}{3}Ma^2$

Uniform disk, radius a , about perpendicular axis through centre: $\frac{1}{2}Ma^2$

Uniform sphere, radius a , about diameter: $\frac{2}{5}Ma^2$

STATISTICS

Summary statistics

$$\bar{x} = \frac{\Sigma x}{n} \quad \text{or} \quad \frac{\Sigma xf}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma (x - \bar{x})^2 f}{\Sigma f}} = \sqrt{\frac{\Sigma x^2 f}{\Sigma f} - \bar{x}^2}$$

Regression and correlation

Regression line of y on x :

$$y - \bar{y} = m(x - \bar{x}), \quad \text{where } m = \frac{n\Sigma xy - \Sigma x\Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

Regression line of x on y :

$$x - \bar{x} = m'(y - \bar{y}), \quad \text{where } m' = \frac{n\Sigma xy - \Sigma x\Sigma y}{n\Sigma y^2 - (\Sigma y)^2}$$

Product moment correlation coefficient:

$$r = \frac{n\Sigma xy - \Sigma x\Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2} \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

Spearman's rank correlation coefficient:

$$\rho = 1 - \frac{6\Sigma d^2}{n(n^2 - 1)}$$

Discrete distributions

$$E(X) = \Sigma xp$$

$$\text{Var}(X) = \Sigma x^2 p - \{E(X)\}^2$$

Continuous distributions

$$E(X) = \int xf(x) dx$$

$$\text{Var}(X) = \int x^2 f(x) dx - \{E(X)\}^2$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

$$F'(x) = f(x)$$

Generating functions

Probability generating function:

$$G_X(t) = E(t^X) = \Sigma p_r t^r$$

$$E(X) = G'(1)$$

$$\text{Var}(X) = G''(1) + G'(1) - \{G'(1)\}^2$$

Moment generating function:

$$M_X(t) = E(e^{tX})$$

$$E(X) = M'(0)$$

$$\text{Var}(X) = M''(0) - \{M'(0)\}^2$$

For $Z = X + Y$, where X and Y are independent:

$$G_Z(t) = G_X(t)G_Y(t)$$

$$M_Z(t) = M_X(t)M_Y(t)$$

Sampling and testing

Unbiased estimators:

$$\hat{\mu} = \frac{\sum x}{n}$$
$$\hat{\sigma}^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Central limit theorem:

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Two-sample tests:

Use of normal distribution: $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

Use of t -distribution: $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$, where $\hat{\sigma}^2 = \frac{\sum(x_1 - \bar{x}_1)^2 + \sum(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$

χ^2 -test:

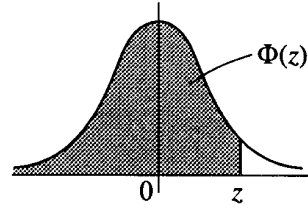
Test statistic is $\sum \frac{(f_o - f_e)^2}{f_e}$

The normal distribution function

If Z has a normal distribution with mean 0 and variance 1 then, for each value of z , the table gives the value of $\Phi(z)$, where

$$\Phi(z) = P(Z \leq z).$$

For negative values of z use $\Phi(-z) = 1 - \Phi(z)$.



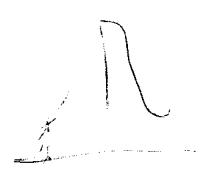
z	0	1	2	3	4	5	6	7	8	9	ADD								
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359	4	8	12	16	20	24	28	32	36
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753	4	8	12	16	20	24	28	32	36
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141	4	8	12	15	19	23	27	31	35
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517	4	7	11	15	19	22	26	30	34
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879	4	7	11	14	18	22	25	29	32
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224	3	7	10	14	17	20	24	27	31
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549	3	7	10	13	16	19	23	26	29
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852	3	6	9	12	15	18	21	24	27
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133	3	5	8	11	14	16	19	22	25
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389	3	5	8	10	13	15	18	20	23
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621	2	5	7	9	12	14	16	19	21
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830	2	4	6	8	10	12	14	16	18
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015	2	4	6	7	9	11	13	15	17
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177	2	3	5	6	8	10	11	13	14
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319	1	3	4	6	7	8	10	11	13
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441	1	2	4	5	6	7	8	10	11
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545	1	2	3	4	5	6	7	8	9
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633	1	2	3	4	4	5	6	7	8
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706	1	1	2	3	4	4	5	6	6
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767	1	1	2	2	3	4	4	5	5
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817	0	1	1	2	2	3	3	4	4
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857	0	1	1	2	2	2	3	3	4
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890	0	1	1	1	2	2	2	3	3
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916	0	1	1	1	1	2	2	2	2
2.4	.9918	.9920	.9922	.9924	.9927	.9929	.9931	.9932	.9934	.9936	0	0	1	1	1	1	1	2	2
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952	0	0	0	1	1	1	1	1	1
2.6	.9953	.9955	.9956	.9957	.9958	.9960	.9961	.9962	.9963	.9964	0	0	0	0	1	1	1	1	1
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974	0	0	0	0	0	1	1	1	1
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981	0	0	0	0	0	0	1	1	1
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986	0	0	0	0	0	0	0	0	0

Critical values for the normal distribution

If Z has a normal distribution with mean 0 and variance 1 then, for each value of p , the table gives the value of z such that

$$P(Z \leq z) = p.$$

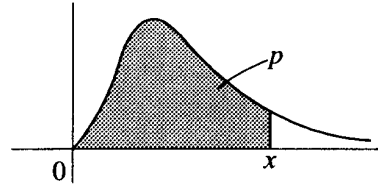
p	.75	.90	.95	.975	.99	.995	.9975	.999	.9995
z	0.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291



Critical values for the χ^2 -distribution

If X has a χ^2 distribution with ν degrees of freedom then for each pair of values of p and ν , the table gives the value of x such that

$$P(X \leq x) = p.$$

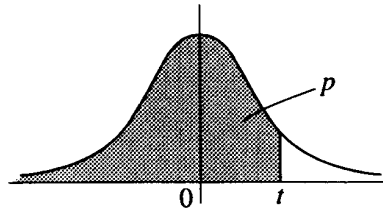


p	.01	.025	.050	.900	.950	.975	.990	.995	.999
$\nu = 1$	0.01571	0.039821	0.03932	2.706	3.841	5.024	6.635	7.879	10.83
2	0.02010	0.05064	0.1026	4.605	5.991	7.378	9.210	10.60	13.82
3	0.1148	0.2158	0.3518	6.251	7.815	9.348	11.34	12.84	16.27
4	0.2971	0.4844	0.7107	7.779	9.488	11.14	13.28	14.86	18.47
5	0.5543	0.8312	1.145	9.236	11.07	12.83	15.09	16.75	20.52
6	0.8721	1.237	1.635	10.64	12.59	14.45	16.81	18.55	22.46
7	1.239	1.690	2.167	12.02	14.07	16.01	18.48	20.28	24.32
8	1.646	2.180	2.733	13.36	15.51	17.53	20.09	21.95	26.12
9	2.088	2.700	3.325	14.68	16.92	19.02	21.67	23.59	27.88
10	2.558	3.247	3.940	15.99	18.31	20.48	23.21	25.19	29.59
11	3.053	3.816	4.575	17.28	19.68	21.92	24.72	26.76	31.26
12	3.571	4.404	5.226	18.55	21.03	23.34	26.22	28.30	32.91
13	4.107	5.009	5.892	19.81	22.36	24.74	27.69	29.82	34.53
14	4.660	5.629	6.571	21.06	23.68	26.12	29.14	31.32	36.12
15	5.229	6.262	7.261	22.31	25.00	27.49	30.58	32.80	37.70
16	5.812	6.908	7.962	23.54	26.30	28.85	32.00	34.27	39.25
17	6.408	7.564	8.672	24.77	27.59	30.19	33.41	35.72	40.79
18	7.015	8.231	9.390	25.99	28.87	31.53	34.81	37.16	42.31
19	7.633	8.907	10.12	27.20	30.14	32.85	36.19	38.58	43.82
20	8.260	9.591	10.85	28.41	31.41	34.17	37.57	40.00	45.31
21	8.897	10.28	11.59	29.62	32.67	35.48	38.93	41.40	46.80
22	9.542	10.98	12.34	30.81	33.92	36.78	40.29	42.80	48.27
23	10.20	11.69	13.09	32.01	35.17	38.08	41.64	44.18	49.73
24	10.86	12.40	13.85	33.20	36.42	39.36	42.98	45.56	51.18
25	11.52	13.12	14.61	34.38	37.65	40.65	44.31	46.93	52.62
30	14.95	16.79	18.49	40.26	43.77	46.98	50.89	53.67	59.70
40	22.16	24.43	26.51	51.81	55.76	59.34	63.69	66.77	73.40
50	29.71	32.36	34.76	63.17	67.50	71.42	76.15	79.49	86.66
60	37.48	40.48	43.19	74.40	79.08	83.30	88.38	91.95	99.61
70	45.44	48.76	51.74	85.53	90.53	95.02	100.4	104.2	112.3
80	53.54	57.15	60.39	96.58	101.9	106.6	112.3	116.3	124.8
90	61.75	65.65	69.13	107.6	113.1	118.1	124.1	128.3	137.2
100	70.06	74.22	77.93	118.5	124.3	129.6	135.8	140.2	149.4

Critical values for the t -distribution

If T has a t -distribution with ν degrees of freedom then, for each pair of values of p and ν , the table gives the value of t such that

$$P(T \leq t) = p.$$



p	.75	.90	.95	.975	.99	.995	.9975	.999	.9995
$\nu = 1$	1.000	3.078	6.314	12.71	31.82	63.66	127.3	318.3	636.6
2	0.816	1.886	2.920	4.303	6.965	9.925	14.09	22.33	31.60
3	0.765	1.638	2.353	3.182	4.541	5.841	7.453	10.21	12.92
4	0.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	0.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	0.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	0.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	0.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	0.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	0.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	0.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	0.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	0.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	0.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	0.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	0.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	0.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	0.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	0.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	0.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	0.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	0.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	0.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.768
24	0.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	0.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	0.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	0.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	0.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	0.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	0.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	0.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	0.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	0.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	0.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

Wilcoxon signed rank test

P is the sum of the ranks corresponding to the positive differences,

Q is the sum of the ranks corresponding to the negative differences,

T is the smaller of P and Q .

For each value of n the table gives the **largest** value of T which will lead to rejection of the null hypothesis at the level of significance indicated.

Critical values of T

One Tail Two Tail	level of significance			
	0.05 0.10	0.025 0.05	0.01 0.02	0.005 0.01
$n = 6$	2	0	–	–
7	3	2	0	–
8	5	3	1	0
9	8	5	3	1
10	10	8	5	3
11	13	10	7	5
12	17	13	9	7
13	21	17	12	9
14	25	21	15	12
15	30	25	19	15
16	35	29	23	19
17	41	34	27	23
18	47	40	32	27
19	53	46	37	32
20	60	52	43	37
21	67	58	49	42
22	75	65	55	48
23	83	73	62	54
24	91	81	69	61